

Topological Schrödinger Bridge Matching

[Yang 2025 ICLR spotlight]

Generative models as distribution matching

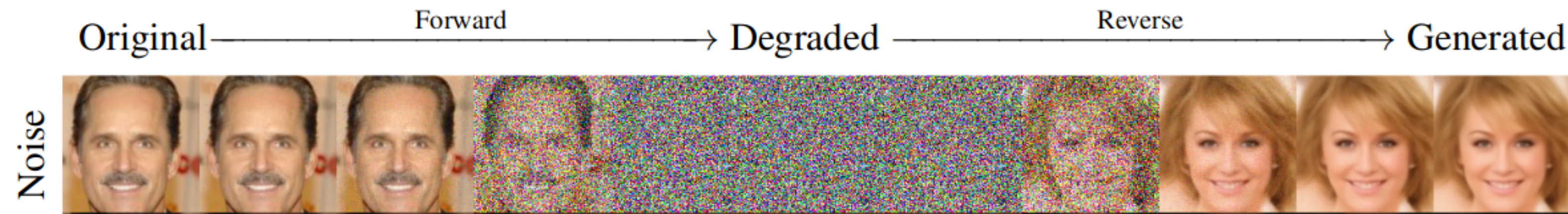


Image generation:
image $\longleftrightarrow$ noise

Forward: image $\rightarrow$ noise

backward: noise $\rightarrow$ image

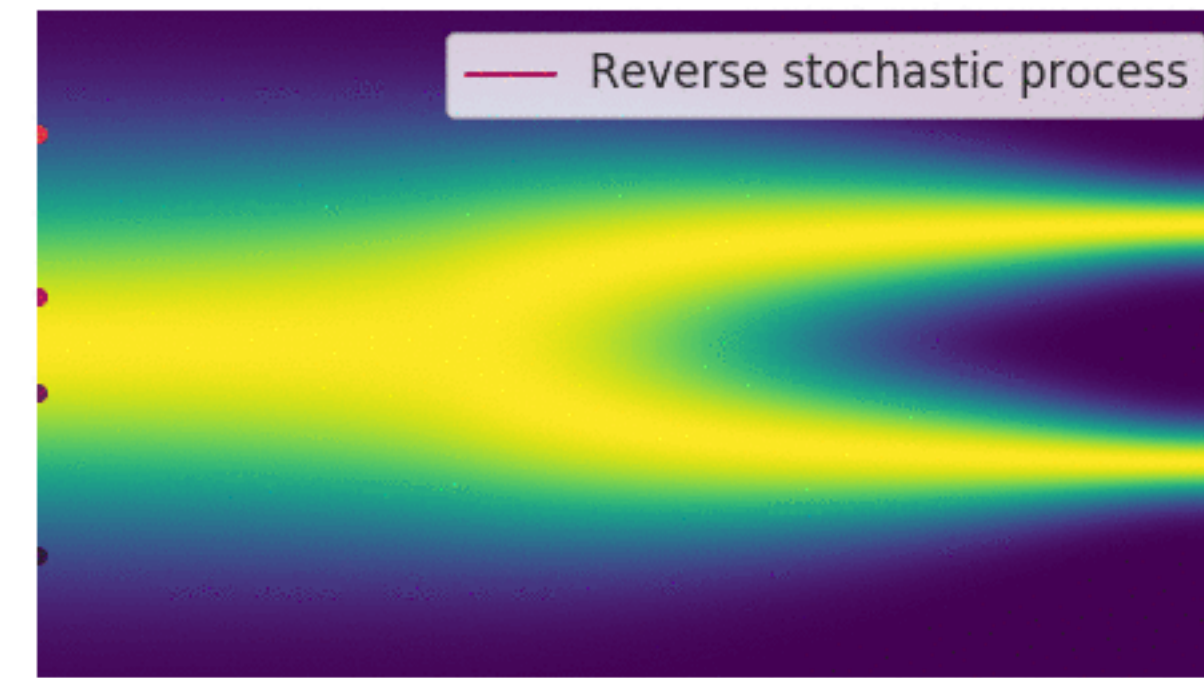
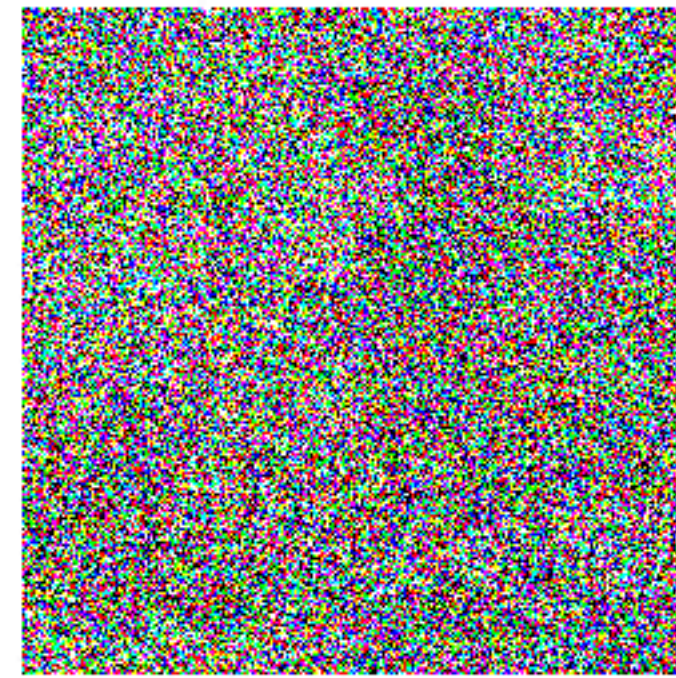
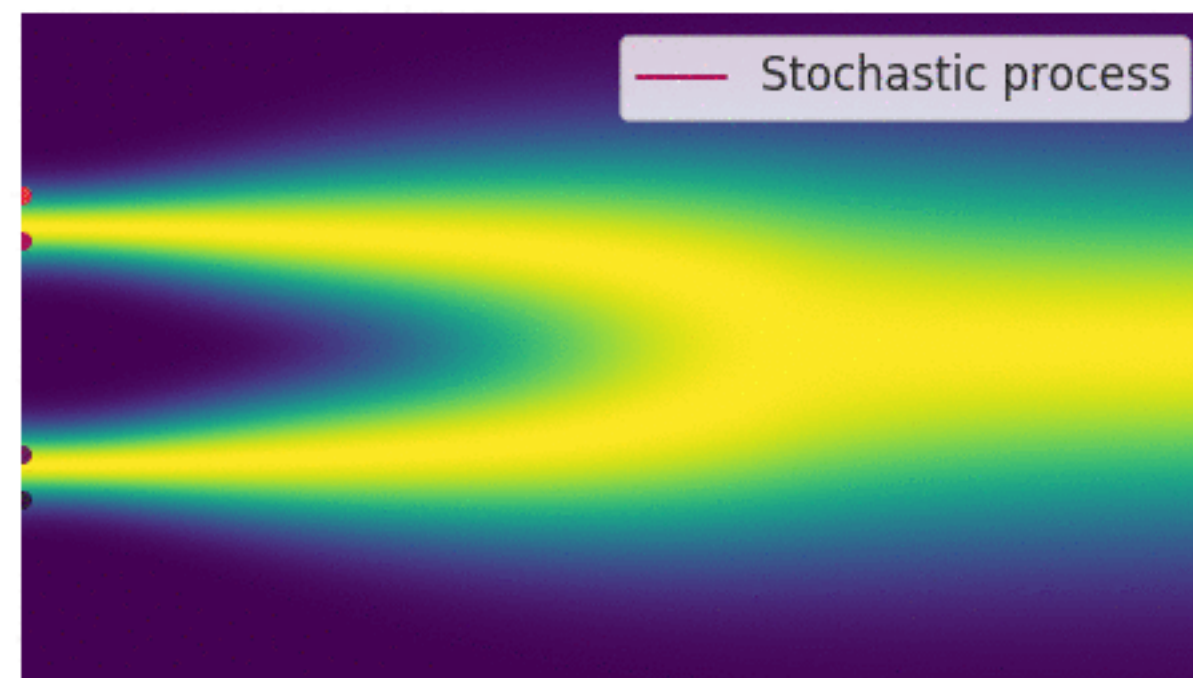


Image matching:
image1 $\longleftrightarrow$ image2



Others:

- VLM: image $\longleftrightarrow$ text
- point clouds 1 $\longleftrightarrow$ clouds 2

[Song et al. 2021]